

Machine Learning-Based Network Intrusion Detection System

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ABSTRACT

The increasing volume of network traffic makes the manual identification of malicious activity increasingly difficult. Intrusion Detection Systems aim to address this problem by identifying suspicious network behavior, while machine learning offers the possibility of learning attack patterns directly from previously labeled traffic.

This project investigates supervised machine learning for network intrusion detection using the CICIDS2017 dataset. A sample of 300,000 network flows, described by 69 numerical features, is used to evaluate both binary intrusion detection and multiclass attack classification. Several models are compared, including Logistic Regression, Random Forest, a tuned Random Forest, and HistGradientBoosting, together with rule-based and majority-class baselines.

HistGradientBoosting achieves the strongest binary performance, with an attack F1-score of approximately 0.995 on a random test split. However, when complete source files are excluded from training, its attack F1-score decreases to approximately 0.755, indicating that generalization to unseen traffic conditions is more difficult. Multiclass classification is also affected by the strong imbalance between attack categories.

Overall, the results demonstrate that machine learning can provide effective intrusion detection, while also showing that realistic evaluation, class imbalance, and model generalization remain important considerations.

1. INTRODUCTION

Modern computer networks continuously exchange large volumes of information between users, devices, servers, and external services. Among this legitimate activity, malicious behavior such as distributed denial-of-service attacks, port scanning, brute-force attempts, botnet activity, web attacks, and infiltration may occur. Detecting such activity manually becomes increasingly difficult as network size and traffic volume grow.

An Intrusion Detection System (IDS) analyzes network activity in order to identify behavior that may correspond to an attack. Traditional approaches commonly use manually defined rules or known attack signatures. These methods can be effective for previously identified patterns, but their performance depends on the quality and completeness of the available rules. Machine learning provides an alternative by learning statistical relationships between network-flow characteristics and known traffic labels.

The objective of this project is to investigate the following research question:

Can machine learning models effectively detect and classify network intrusion attempts from traffic-flow data, and how do they compare with simpler rule-based detection approaches?

To answer this question, a complete machine-learning pipeline is implemented using the CICIDS2017 dataset. The project considers two tasks. The first is **binary intrusion detection**, in which each flow is classified as either BENIGN or ATTACK. The second is **multiclass classification**, in which the objective is to identify the specific type of attack.

A major challenge is that the dataset is imbalanced, since benign traffic is considerably more frequent than malicious traffic and some attack categories contain very few observations. For this reason, the evaluation focuses particularly on precision, recall, and F1-score for malicious traffic rather than relying only on overall accuracy.

The project compares several supervised classifiers with simple baselines, evaluates model stability and generalization, and investigates which network-flow features contribute most strongly to the predictions.

2. RELATED WORK

Machine learning has been widely investigated for intrusion detection because network-flow data contain measurable statistical characteristics that can differ between normal and malicious activity. An important requirement for this research is the availability of labeled traffic containing both realistic benign activity and representative cyberattacks.

CICIDS2017 provides network traffic collected across several days and includes normal activity together with multiple attack scenarios [1]. Its attack categories include denial-of-service attacks, DDoS, brute-force activity, botnet traffic, port scanning, web attacks, infiltration, and Heartbleed. Each network flow is represented using numerical characteristics such as packet sizes, flow duration, packet timing, TCP flags, and forward and backward communication statistics.

Tree-based ensemble methods are particularly appropriate for this type of structured numerical data because they can represent nonlinear relationships between multiple features. Random Forest combines several decision trees, while gradient-boosting methods construct trees sequentially to correct errors made by previous stages.

However, strong results on a random train-test split do not necessarily guarantee equivalent performance on traffic collected under different conditions. For this reason, the project also includes a source-file holdout evaluation in addition to the standard random split. Feature-importance and SHAP-based interpretation are further used to provide insight into the characteristics influencing model predictions.

3. SCIENTIFIC BACKGROUND

3.1 Network Intrusion Detection

Network intrusion detection consists of analyzing network activity in order to identify potentially malicious behavior. In this project, traffic is represented as **network flows**, where each observation summarizes communication through numerical properties such as packet counts, packet sizes, flow duration, inter-arrival times, TCP flags, and forward and backward traffic statistics.

Two classification settings are considered. In **binary classification**, all malicious traffic is grouped under the ATTACK class while legitimate traffic remains BENIGN. In **multiclass classification**, the original attack categories are retained so that the model attempts to identify the specific type of malicious activity.

Binary classification provides the main security decision of whether a flow should be considered suspicious. Multiclass classification provides more detailed information but is more challenging because many individual attack categories contain substantially fewer observations.

3.2 Supervised Machine Learning

Three main types of supervised models are evaluated. **Logistic Regression** provides a relatively simple linear reference classifier. **Random Forest** combines several decision trees and can model nonlinear relationships between network-flow features [2]. A tuned Random Forest is also evaluated using optimized hyperparameters. Finally, **HistGradientBoosting** builds decision trees sequentially and is designed for efficient learning on structured numerical datasets [3].

The machine-learning approaches are compared with two simpler baselines. A **Dummy Majority** classifier always predicts the most frequent class, while a **Rule-Based baseline** uses manually selected conditions to identify potentially malicious traffic.

3.3 Evaluation Measures

The models are mainly evaluated using precision, recall, and F1-score. **Precision** measures how often predicted attacks are actually malicious, while **recall** measures how many real attacks are successfully detected. **F1-score** combines precision and recall and is particularly useful when the dataset is imbalanced [5].

Accuracy is also reported, but it must be interpreted carefully. Since most observations are benign, a classifier can obtain relatively high accuracy by predicting BENIGN frequently while still failing to detect attacks.

The binary dataset contains approximately 84.84% benign traffic and 15.16% attack traffic, corresponding to an imbalance ratio of approximately 5.6:1. The multiclass problem is even more imbalanced because some attack categories contain thousands of observations while others contain only a small number. This imbalance strongly affects the ability of a classifier to recognize rare attacks.

4. METHODOLOGY

4.1 Dataset and Sampling

The experiments use the CICIDS2017 network-flow dataset, loaded from eight CSV files corresponding to different traffic collection periods. The complete dataset contains approximately 2.8 million flows. To reduce memory usage and training time during development, a sample of **300,000 flows** is used.

After preprocessing, the dataset contains **69 numerical features** and **15 original labels**, including BENIGN and fourteen attack categories.

Property	Count
Sampled Network Flows	300 000
Numerical Features	69
Original Labels	15

Table 1. Dataset Overview

Binary Label	Flows	Percentage
BENIGN	254 521	84.84%
ATTACK	45 479	15.16%

Table 2. Binary Class Distribution

4.2 Training and Evaluation

Logistic Regression, Random Forest, and HistGradientBoosting are trained and compared with the rule-based and majority-class baselines. A tuned Random Forest is also produced using hyperparameter search, with the best tested configuration using 120 trees, a maximum depth of 24, and a minimum leaf size of one.

Three-fold stratified cross-validation is used to determine whether model performance remains consistent across different subsets of the training data.

In addition to the random split, a **source-file holdout** experiment is performed. Complete source CSV files are reserved for testing rather than having their flows randomly distributed between the training and testing sets. This produces 224,964 training observations and 75,036 test observations and provides a more demanding evaluation of how well the models

generalize to traffic sources that were not observed during training.

Finally, feature-importance analysis and SHAP are used to investigate which network-flow characteristics influence predictions made by the tree-based classifiers.

5. EXPERIMENTAL RESULTS

5.1 Binary Intrusion Detection

The strongest results are obtained using tree-based models. HistGradientBoosting achieves the best attack F1-score at approximately **0.9953**, followed closely by the tuned Random Forest at **0.9944** and the original Random Forest at **0.9940**.

Logistic Regression obtains a lower attack F1-score of approximately **0.8047**. Its recall remains high at 0.9560, but its lower precision indicates that it produces considerably more false-positive alerts than the tree-based models.

The Rule-Based baseline achieves an attack F1-score of approximately **0.4162**. The Dummy Majority classifier reaches an overall accuracy of approximately 0.8484 but has an attack F1-score of zero because it always predicts BENIGN. This illustrates why accuracy alone is not sufficient for evaluating intrusion detection.

Model	Accuracy	Attack Precision	Attack Recall	Attack F1
HistGradientBoosting	0.9986	0.9948	0.9957	0.9953
Tuned Random Forest	0.9983	0.9954	0.9933	0.9944
Random Forest	0.9982	0.9947	0.9932	0.9940
Logistic Regression	0.9297	0.6948	0.9560	0.8047
Rule-Based Baseline	0.8611	0.5738	0.3266	0.4162
Dummy Majority	0.8484	0.0000	0.0000	0.0000

Table 3. Main Binary Model Comparison

Error analysis of HistGradientBoosting also shows that only **59 benign flows are incorrectly classified as attacks**, while **49 attacks are incorrectly classified as benign** in the random test set. Although

both error types are limited, false negatives remain particularly important because they represent attacks that would not generate an IDS alert.

Model	Mean Attack F1	Standard Deviation
HistGradientBoosting	0.9917	0.0013
Random Forest	0.9890	0.0022
Logistic Regression	0.8076	0.0034

Table 4. Three-fold Cross-Validation Results

Three-fold cross-validation confirms the stability of the strongest models. HistGradientBoosting achieved the highest mean attack F1-score of 0.9917, followed closely by Random Forest at 0.9890. The low standard deviations indicate that both tree-based models performed consistently across the three folds.

5.2 Source-File Holdout

The source-file holdout experiment produces a substantially more difficult classification problem.

Model	Accuracy	Attack Precision	Attack Recall	Attack F1
HistGradientBoosting	0.8955	0.9964	0.6079	0.7551
Random Forest	0.8952	0.9947	0.6079	0.7546
Logistic Regression	0.8659	0.8378	0.6131	0.7080

Table 5. Source-File Holdout Results

HistGradientBoosting remains the strongest model, but its attack F1-score decreases from approximately 0.9953 on the random split to 0.7551. An important observation is that HistGradientBoosting maintains extremely high precision, approximately 0.9964, while recall decreases to approximately 0.6079. Therefore, when the model predicts ATTACK, the prediction is usually correct, but a considerably larger proportion of attacks remains undetected.

This result demonstrates that a random train-test split can provide a more optimistic estimate of intrusion-detection performance. Network flows originating from the same or similar collection conditions may share characteristics that make classification easier.

Testing on completely unseen traffic sources therefore provides a more realistic indication of model generalization.

5.3 Multiclass Classification

The multiclass experiment attempts to identify the exact traffic category rather than only separating benign and malicious flows.

HistGradientBoosting reaches an overall multiclass accuracy of approximately **0.83** and a weighted F1-score of approximately **0.8451**. However, its macro F1-score is only approximately **0.1475**.

The difference between weighted and macro F1 reflects the strong class imbalance. The model classifies frequent categories such as BENIGN and DDoS considerably more successfully than rare attack categories. DDoS reaches an F1-score of approximately **0.96**, while PortScan reaches approximately **0.32**. Several rare categories receive scores close to zero because only a limited number of examples are available.

These findings show that strong binary intrusion detection does not automatically imply equally reliable identification of every specific attack type.

5.4 Feature Interpretation

Feature analysis indicates that packet-length and TCP window characteristics play an important role in distinguishing malicious and benign traffic. Important variables include `Fwd_Packet_Length_Max`, `Init_Win_bytes_forward`, `Init_Win_bytes_backward`, average forward packet length, average forward segment size, and packet-length variability.

Permutation analysis also identifies `Init_Win_bytes_forward` as particularly influential. SHAP analysis is applied to HistGradientBoosting to provide additional information about how individual features affect model predictions [4].

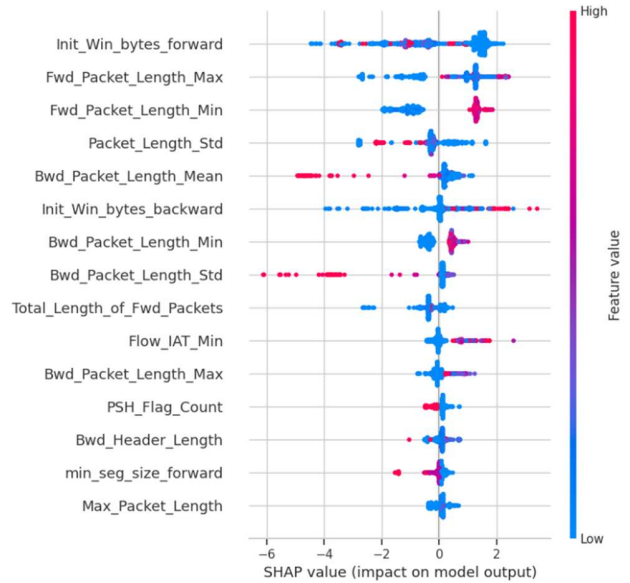


Figure 1. SHAP Feature Impact for the HistGradientBoosting Model

The explainability analysis supports the observation that the model uses multiple network-flow characteristics rather than depending entirely on a single feature. This provides additional transparency when interpreting the decisions of the best-performing classifier.

6. CONCLUSION

This project investigated whether supervised machine learning can effectively detect and classify network intrusions using traffic-flow data from CICIDS2017. The results show that machine-learning classifiers substantially outperform the simplified rule-based and majority-class baselines for binary intrusion detection.

HistGradientBoosting achieves the strongest binary result, with an attack F1-score of approximately **0.9953** on the random test split. Random Forest performs at a very similar level, and cross-validation confirms that both tree-based approaches remain stable across different data partitions. These results demonstrate that network-flow characteristics contain strong statistical patterns that can be used to distinguish benign and malicious traffic.

However, the source-file holdout experiment provides an important qualification to these results. When traffic from complete source files is excluded from training, the best attack F1-score decreases to approximately **0.7551**, mainly because attack recall falls considerably. This indicates that performance on randomly separated data does not necessarily represent performance on traffic collected under different conditions.

Multiclass classification also remains challenging because rare attack categories contain too few examples for reliable learning. Overall, machine learning is therefore highly promising for intrusion detection, but realistic evaluation and the ability to generalize beyond the training distribution are essential before such systems are considered for practical deployment.

7. LIMITATIONS AND FUTURE WORK

Several aspects of the current system could be improved.

First, the experiments use a sample of 300,000 observations rather than the complete CICIDS2017 dataset. Future work could train the strongest classifiers on the full dataset and investigate improved techniques for handling rare attack categories. A two-stage approach could also be explored, where one model first detects malicious traffic and a second model identifies the specific attack type.

Second, stronger evaluation strategies could be used. The source-file holdout experiment already demonstrates that performance decreases when the test traffic differs from the training data. Future experiments could therefore use stricter temporal or source-separated splits in which models are trained on earlier traffic and evaluated on subsequent collection periods.

Finally, the current implementation operates offline using previously collected CSV files. A future version could process live network flows and generate alerts in real time. Explainability methods such as SHAP

could also be integrated into this prototype so that security analysts receive both an intrusion prediction and information about the network characteristics that contributed to the decision.

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